

AZ-104 Lab 09c – Implement Azure Container Apps

Lab Objective

To understand how to deploy and run containerized applications using **Azure Container Apps (ACA)**, a **serverless container platform** built on top of managed Kubernetes, without directly managing clusters or infrastructure.

Core Concepts Explained (Important)

◆ What is Azure Container Apps (ACA)?

Azure Container Apps is a **serverless platform for containerized applications** that:

- Uses Kubernetes internally
- Abstracts all Kubernetes complexity
- Supports autoscaling
- Supports ingress, revisions, and microservices patterns

You **do not manage**:

- Nodes
- Pods
- Kubernetes control plane
- Networking or ingress controllers

Azure manages everything.

◆ Container Apps Environment

A **Container Apps Environment**:

- Is a **logical boundary** for container apps
- Hosts the **managed Kubernetes cluster**
- Can contain **multiple container apps**
- Is required before creating any container app

Think of it as:

“A managed Kubernetes workspace”

◆ When to Use Azure Container Apps

Use ACA when:

- You want Kubernetes features
 - You want autoscaling
 - You want ingress + revisions
 - You don't want to manage AKS
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Task 1 – Create Azure Container App & Environment

What You Did

1. Navigated to **Azure Portal** → **Container Apps**
 2. Selected **Create** → **Container App**
 3. Configured:
 - Resource Group: az104-rg9
 - Container App Name: my-app
 - Region: East US
 4. Created a new **Container Apps Environment**
 - Environment Name: my-environment
 5. Selected **Quickstart Image**
 - Simple Hello World container
 6. Reviewed and created the app
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What Happened Behind the Scenes

Azure automatically:

- Created a managed Kubernetes cluster
- Configured ingress
- Assigned a public DNS name

- Deployed the container
- Enabled scaling

All without user interaction.

Task 2 – Test & Verify Deployment

Verification Steps

1. Navigated to the container app resource
 2. Copied **Application URL**
 3. Opened the URL in browser
 4. Verified message:
 5. Your Azure Container Apps app is live
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Why This Matters

This confirms:

- Container is running
 - Ingress routing is working
 - Public DNS is correctly mapped
 - Application is accessible
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Cleanup

Deleted the resource group to:

- Stop billing
 - Avoid unnecessary costs
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Key Learnings (Lab 09c)

- Azure Container Apps abstracts Kubernetes complexity
- Ideal for long-running containerized services
- Supports autoscaling and ingress

- Scale-to-zero reduces cost
 - Best alternative to AKS for small to medium apps
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Interview Questions & Answers – Lab 09c

Q1: What is Azure Container Apps?

A serverless container platform built on managed Kubernetes.

Q2: Does ACA use Kubernetes?

Yes, but it is fully managed and abstracted.

Q3: Difference between ACA and AKS?

ACA removes cluster management; AKS requires full Kubernetes management.

Q4: Can ACA autoscale?

Yes, including scale-to-zero.

AZ-104 LAB 09 – FULL WRAP-UP (a + b + c)

Lab 09a – Azure Web Apps

Feature	Description
Service Type	PaaS
Deployment	Code-based
Scaling	Manual / Autoscale
Best For	Traditional web apps
Infrastructure	Fully managed

Lab 09b – Azure Container Instances (ACI)

Feature	Description
Service Type	Serverless Containers

Feature	Description
Deployment	Single container
Scaling	No autoscaling
Best For	Short-lived jobs
Kubernetes	✗ No

◆ Lab 09c – Azure Container Apps (ACA)

Feature	Description
Service Type	Serverless Kubernetes
Deployment	Containerized apps
Scaling	Autoscale + scale-to-zero
Best For	APIs, microservices
Kubernetes	✓ Managed

🔥 One-Line Exam Summary

ACI = Simple containers,
ACA = Kubernetes without complexity,
AKS = Full Kubernetes control